S(15): COMPLETE DETERMINATION WITH EXACT COMPUTER ASSISTANCE

29 October 2026

Theorem. For arbitrary elliptic curves over number fields of exact degree15,

$$S(15)=\{2,3,5,7,11,13,17,19,23,29,31,37,43\}.$$

The last candidate61 is excluded by the genus-8 quotient argument in Section8. All primes through the dyadic bound33131 are accounted for.

1. OCCURRENCE AND THE DEGREE CONDITION

The published classification S(5)=Primes(19) gives all primes through19: extend a quintic field by a cubic extension. The resulting field has exact degree15 and retains the torsion point.

For each p=23,29,31,37,43, occurrences.json records a degree15 irreducible polynomial f, and functions x=a,y=g(a)/h(a), with f(a)=0. Set

$$r=(x^2*y-x*y+y-1)/(x^2*y-x),\quad s=(x*y-y+1)/(x*y),$$
$$b=r*s*(r-1),\quad c=s*(r-1).$$

Then E=[1-c,-b,-b,0,0], P=(0,0), supplies an occurrence. verify_occurrences.m checks irreducibility over Q, nonsingularity, P nonzero and p*P=0. These are exact degree15 number fields. The examples at37 and43 come from van Hoeij's low-degree-place table; the other three are explicitly verified specializations.

For exclusion, any prime dividing E(K)_tors is the order of a K-rational point. Its noncuspidal point on X_1(p) has degree dividing15. For p>37, proper divisors1,3,5 are excluded by the known classifications over Q, cubic fields and quintic fields. Thus it has exact degree15. Write D for its irreducible rational trace divisor on X_1(p).

2. THE DYADIC GLOBAL BOUND

We use the previously established dyadic doubling argument, including the cotangent-lattice defect. Its complete statement and references are included in prior_dyadic_theorem/asymptotic_prime_bound.tex. For J=J_0(p), integral Hecke algebra T, winding element e and I=Ann_T(e), let A=J/IJ be the optimal winding quotient. Then A(Q) is finite. If

$$M=H^0(X_0(p)_(Z_2),\Omega^1)[I],\quad L=\pi^*\mathrm{Cot}(A_(Z_2)),$$

Raynaud's theorem gives 2M subset L subset M. Equality is not assumed.

Use m=30=2*15 coefficients and M0=157. Write

$$T'_n=\sum_{a|n}\mu(n/a)T_a,\quad L_n=T'_{2n}-2*T'_n.$$

The spans of L_1,...,L_30 and (T_2-3)T_1,...,(T_2-3)T_30 coincide. For p>2*m*M0=9420 and p*u=1 modulo157, the DKSS intersection formula for L_n e against {0,a/157} is

$$\varepsilon(na)-\varepsilon(nu/a),\quad 1\le n\le 30,\ 1\le a\le 156,$$

where epsilon indicates the upper half of the nonzero residues mod157. For each of the156 values u, the resulting30 by156 matrix has rank30 modulo2. The independent Sage checker rebuilds all entries and ranks.

These intersection matrices prove surjectivity of the first30 coefficient map on M. Thus the cotangent image of the Abel map

$$X_0(p)^{(30)} \to A$$

at30*infinity contains2*Z_2^30. The local lattice lemma in the cited prior theorem then proves injectivity in the formal neighbourhood modulo4: a nonzero minimum-valuation coordinate of valuation k>=2 has a linear term of valuation k+1<2k.

If a degree15 divisor reduces to15*c modulo2, its elementary symmetric polynomial is1+2B(t). Doubling the divisor squares that polynomial, giving1+4(B+B^2), so the double agrees with30*c modulo4. The torsion reduction kernel of an abelian scheme over Z_2 is killed by2. Consequently a rational degree15 divisor with entirely cuspidal reduction of the indicated type is impossible on the winding quotient.

If p>(2^(15/2)+1)^2, reduction of E at every place above2 cannot be good, by Hasse. Additive reduction cannot support this prime torsion. The nonsplit multiplicative and multiplicative identity-component bounds also fail. The point therefore lies in the component group in split multiplicative reduction. After Atkin--Lehner on X_0(p), its trace reduces to15*infinity. The preceding doubled-divisor argument excludes it. Since floor((2^(15/2)+1)^2)=33131 and9420<33131, we obtain

p<=33131.

Using30 coefficients, rather than15, is essential here.

3. PUBLISHED EXCLUSIONS

We apply Derickx--Stoll, arXiv:2505.14109v1, Table2, Proposition4.1, Remark4.4 and Corollary7.5, only within their stated range p<100000. With C=2^19/325, Corollary7.5 says that for p in S_new(15), either p^2<=15*C+1, or (p-1)/(2*str(p)) divides15 and there is a noncuspidal point of the specified degree on the str(p)-fold cover of X_0(p). Its two CM alternatives require even degree and do not apply. The exact arithmetic ledger checks this at every prime up to33131.

For a non-strange prime, (h-1)J_1(p)(Q) is finite for every diamond h. Proposition4.1 with ell=2,a=2,n=5 excludes further primes when the bound325*(p^2-1)/65536 exceeds75 and no possible shortened diamond orbit length divides15. Remark4.4 supplies the rational2-torsion hypothesis for these levels.

At151, take a generator of the full diamond group, of order75, and n=6. The gonality bound exceeds90=6*15. Its noncuspidal orbits in characteristic zero have length75, or25 at j=0; neither fits in an effective divisor of degree15. There is no order2 shortening of an odd-order group. This excludes151.

At181, str(181)=6 and Corollary7.5 requires a noncuspidal rational point on the degree6 intermediate cover of X_0(181). Its image would be a rational noncuspidal point on X_0(181), contrary to Mazur’s prime-isogeny theorem. This excludes181.

Apart from the occurrence43, the levels left for local analysis are
41,47,53,59,61,67,71,73,79,83,89,97,101,103,107,109,113.

4. CUSPIDAL REDUCTIONS AND FINITE-IMAGE CERTIFICATES

At these17 levels, actual integral endomorphisms with finite rational image are composed with an odd-prime torsion annihilator T_q-<q>-q in the mu-level convention. The resulting maps kill the whole rational Jacobian group. Fricke transports the formal-immersion criterion to the geometric point-level rational cusps.

The chosen intermediate-curve diamond indices are
41:20,47:23,53:13,59:29,61:15,67:11,71:5,73:12,79:39,
83:41,89:11,97:8,101:5,103:17,107:53,109:54,113:56.

All normalized cusp multiplicity patterns of degree15 pass. The core uses15 rows at the first cusp and6 at each other cusp. The only patterns requiring a seventh row at another cusp are8+7 and7+7+1; all relative choices of cusps are checked separately. The core is verified by complete prefix enumeration or by spanning relation bases and a complete binary search through their relation space. Independent dynamic-programming counts check every prefix enumeration. Sage independently verifies the full matrices, exceptional patterns and relation bases. All17 selected certificates pass.

The finite-image claims do not rely on a winding-vector computation in an incompatible basis. The exported successful polynomials are fixed explicitly in finite_FI_polynomials.m. At61,97,101, the selected indices are respectively not divisible by6,12,10, so the intermediate Jacobians contain no nontrivial strange character. All positive-rank factors of these and of the other non-strange intermediate Jacobians therefore lie in their trivial-character part.

verify_finite_FI_final.m verifies all16 selected nontrivial polynomial inclusions on the small Gamma_0 spaces. A projected winding row transforms by TRANSPOSE(DualHeckeOperator), which is the correct period-coordinate action. The span of its T_2-powers is a subspace of the winding module. For each selected polynomial the script proves that its operator image is contained in this span. Independent Sage rational linear algebra checks the inclusions again. At41,47,59,71 the whole Jacobian has rank zero; at107,109,113 a diamond difference kills all positive-rank factors. Thus the claimed finite images are justified independently of the heuristic initially used to choose them.

5. FINITE-FIELD CENSUS AND ORDINARY REDUCTIONS

The census enumerates all j-invariants and twists over F_(2^d) for 1<=d<=15. It enumerates every cyclic p-subgroup when the rational p-torsion has rank2, rather than assuming it is always cyclic. Tate coordinates determine the exact closed-point degree and full closed diamond orbit. Frobenius-invariant keys remove duplications. It finds2175 ordinary source orbits at the17 levels.

The relevant supersingular types (degree, closed diamond orbit size) are
41:(5,2);53:(13,1);61:(15,1);73:(6,4);
109:(9,3),(12,3);113:(7,4).
The nonrational cusp degrees that fit are
41:10,73:9,89:11,113:14.

In the geometric convention delta_a(E,P)=(E,aP), put
A3=T_3-1-3*delta_3.
It kills rational torsion and each rational cusp divisor. For ordinary supports we use alpha*A3, with finite-image alpha as follows:
41,47,59,71: alpha=1;
61: alpha=sigma^6-1;
97: alpha=sigma^12-1, with (sigma-1)*Phi12(sigma) on the three degree10/14 closed orbits of size24;
101: alpha=(sigma-1)*(sigma^5+1);
all other levels: alpha=sigma-1.
Here sigma is the recorded primitive diamond, with the inverse point convention specified in the input. Character orders are unchanged by inversion. At97 the four-term polynomial above kills character orders1 and12 and keeps the support small enough for the canonical sections.

For a degree-zero effective difference U-V that would have to be principal, choose R=H^0(K^2) at41,47,53 and R=H^0(K) elsewhere. The calculation constructs actual Hecke targets and local canonical jets from elliptic curves and torsion points. Old jets are reused only after a unique exact matching against the fresh census. Tensor products of sections are verified by q-expansions beyond the applicable Sturm bound. At109 and113 the variable precision is sufficient for the universal degree120 support bound; requested multiplicities are checked.

If div(f)=U-V, then for any F,H in R(-U), the section S=F/f lies in R(-V) and H*S is divisible by F in the product section space. Thus the homogeneous linear conditions on S have a nonzero solution. A full-rank matrix proves nonprincipality. This implication holds even if the chosen F also vanishes at some negative support points.

There are2173 independently verified full-rank ordinary matrices.
The only ordinary exceptions are
 41: degree14, orbit10;
 113: degree8, orbit7.
The former is treated in Section6; the latter by a stabilizer in Section7.

Two ordinary points can occur only with degrees7+7 or7+8. All137 7+7 orbit configurations from the degree14 computation remain excluded with an extra rational cusp. All254 new7+8 configurations pass:
 59:58,61:90,67:66,71:40.
The only mixed supersingular/ordinary possibilities are
 41:5+8,5+9,5+10;73:6+7;113:7+8.
Using (sigma^2-1)A3 at41 gives13 full-rank matrices; using (sigma^4-1)A3 at73 gives one. These operators kill the corresponding supersingular contributions and all rational cusp contributions.
The113 mixed case is treated in Section7.
Sage independently checks all2578 ordinary, pair and mixed matrices, including the137 borrowed7+7 matrices.

6. THE EXCEPTIONAL LEVEL41 CONFIGURATIONS

There are two supersingular degree5 points, interchanged by diamonds. The previously independently audited native Magma certificates exclude one such point, both possibilities for a pair (including repetition), and a nonrational degree10 cusp. They remain valid with any rational cusp complement, because A3 kills that complement.

exceptional41.m adds the two triple types3P and2P+Q. Both A3 images are nonprincipal. It also excludes a degree10 nonrational cusp plus either degree5 supersingular point: delta_3 swaps the two nonrational cusps, so the image is2(C1-C2)+A3(P). Both native tests pass.
These cover all new purely supersingular/cuspidal configurations.

The surviving ordinary degree14 orbit consists of reductions of known characteristic-zero degree14 points on X_1(41). lift41_charzero.m checks an exact irreducible number-field example; known41_lifting.m directly matches its Tate coordinates to the finite-field source. Diamonds supply a characteristic-zero lift for every closed point in the orbit.

For this degree14 divisor plus each of the20 rational cusps, the joint cotangent matrices of the actual q=3,5,7 torsion-annihilating maps have rank15. Sage independently checks all20 matrices. Formal immersion makes a degree15 rational divisor with this reduction equal to the known reducible lift B14+c. That contradicts irreducibility of D.

This resolves the degree14 ordinary exception at41.

7. STABILIZER EXCLUSIONS

We use the following elementary lemma. Suppose h fixes the nonrational
part of the reduction of D, and C is the rational cusp lift of the
remaining part, of degree c. If (h-1)J(Q) is finite, then
 y=[h(D-C)-(D-C)] is rational torsion reducing to zero, so2y=0.
If y=0 and gon_Q(X)>15+c, or merely2y=0 and gon_Q(X)>2*(15+c),
there can be no nonconstant rational function with that divisor.
Consequently hD=D and hC=C. Diamonds act freely on rational cusps,
so the order of h must divide c.

For an odd-order h acting trivially on J(Q)[2], the norm relation
(1+h+...+h^(r-1))y=0 gives y=0. The actual JOne homology certificates
show J_1(53)(Q)[2]=0, and diamond2 acts trivially on J_1(73)(Q)[2].
Their integral lattices are those of JOne, not an isogenous substitute.

At53 the sole supersingular degree13 point has complement c=2 and is
fixed by the order13 diamond4. We prove gon_Q(X_1(53))>=18, which
exceeds17. Hence13 would divide2, a contradiction. For the gonality
bound, the curve has26 rational points and no other places through
degree8. If a function has degree<=17, one of its three F_2 fibres
contains at least9 distinct rational points. Move that fibre to infinity.
All its poles must then be rational, and their divisor is bounded by
C_all+E, where E has degree8. Exact canonical evaluation gives rank26
at C_all and8 additional independent conditions for every normalized E.
All1,143,531 configurations are checked, against an independent count.
Riemann--Roch gives l(C_all+E)=1, excluding the function.

At73 use the order9 diamond2 and gonality>26. The complements for
one supersingular point, two supersingular points, and one nonrational
cusp are respectively9,3,6. The latter two fail9|c. For the first,
hD=D forces every geometric h-orbit in the irreducible degree15 divisor
to have length3, since9 does not divide15. This forces j=0 and a
nontrivial order3 elliptic automorphism preserving the p-line. It is
defined over the residue field: its nontrivial conjugate sends the
nonzero p-point to a different scalar multiple. Thus that degree15
field contains Q(sqrt(-3)), impossible. The same argument excludes a
supersingular degree6 point plus a degree9 nonrational cusp (c=0).

The remaining stabilizer comparisons use the doubled bound:
 level, support degree, c, diamond, order, required gonality bound
 89, nonrational cusp11,4,2,11, >38;
 109, supersingular9,6,23,18, >42;
 109, supersingular12,3,23,18, >36;
 113, ordinary8 orbit7,7,40,8, >44;
 113, supersingular7,8,2,14, >46;
 113, two supersingular7,1,2,14, >32;
 113, nonrational cusp14,1,2,14, >32.
In each case325*(p^2-1)/65536 exceeds the required number, and the
listed order does not divide c. Exact Tate-coordinate computations
verify every asserted closed-point stabilizer, and diamonds commute
with their translates.

For113 the mixed degree7 supersingular plus degree8 ordinary case has
c=0. The unique diamond involution, represented by98=3^28 mod113,
fixes both closed points, for both ordinary orbits7 and28. Gonality>30
forces hD=D. An involution permuting an irreducible odd-degree divisor

must fix each point, since all its orbit lengths are equal. A nontrivial diamond involution can fix a characteristic-zero noncuspidal point only at j=1728, with the p-line preserved by the order4 automorphism. The same conjugation argument forces Q(i) into the degree15 field, impossible.

These arguments and Section6 cover all supersingular and nonrational cusp configurations at levels other than61.

8. EXCLUDING61 BY THE GENUS-8 QUOTIENT

The rational-cusp cases and all149 single ordinary source orbits at61 pass. All16 degree7+7 pair configurations with a rational cusp and all90 degree7+8 configurations also pass. The only remaining reduction for an irreducible degree15 divisor D on X=X_1(61) is the unique supersingular closed point P15 of degree15, whose closed diamond orbit has size1. This section excludes that reduction.

Write delta_a for (E,P) -> (E,aP), and put
 Delta=(Z/61Z)^*/{+1,-1},
 H5=<delta_3>=<delta_2^6>, H15=<delta_4>,
 Y=X/H5, Z=X/H15, C=X_0(61).
The successive degrees in X -> Y -> Z -> C are5,3,2. Their genera are respectively126,26,8,4. Let s be the involution of Z induced by delta_2. Thus Z/<s>=C. All curves, maps and diamond actions are defined over Q. The new scripts and logs are in work/degree15/quotient61; a separate proof with replay instructions is provided there in PROOF61.txt.

8.1. EXACT JACOBIAN AND REDUCTION CERTIFICATES

The following properties are verified by the supplied calculations:
 (i) (delta_3-1)J_X(Q) is finite;
 (ii) (s-1)J_Z(Q) is finite;
 (iii) J_X(Q)[2]=0;
 (iv) delta_2 and delta_3 fix P15 as a closed divisor.

For (i) and (ii), finite_diamonds.m works in the sign+ cuspidal modular symbol spaces of dimensions126 and8. In projected period coordinates, the Hecke action is the transpose of DualHeckeOperator. The script checks compatibility with the ambient Hecke action through the exact projection matrix; it does not identify matrices in different bases. It recovers D2=(T2^2-T4)/2 and uses the projected winding vector e. The rows e*T2^k, 0<=k<g, span spaces of dimensions123 and7. The images of D2^6-1 and D2-1, respectively of dimensions100 and4, lie in these winding spaces. This is checked by rational matrix ranks. The winding rank-zero theorem therefore gives (i) and (ii). Inverting the diamond convention changes neither the relevant kernels nor their images.

For (iii), two61_data.m contains integral matrices on the actual sign0 Betti homology lattice of JOne(61), of rank252. For q=3,5,7, the matrices A_q=T_q-1-q*<q> annihilate rational2-torsion, by good reduction and Eichler--Shimura. Their common kernel modulo2 has dimension0. This calculation uses the actual Jacobian lattice, not a Jacobian that is merely isogenous to it. The generating script is work/audit11/rational_two.m; the original log is work/degree15/two61.log.

For (iv), verify_reduction.m reconstructs the actual elliptic curve and61-point representing P15. Its Tate coordinates have degree15. On these coordinates delta_2, delta_3 and delta_4 act as Frobenius to the powers2,12 and4, respectively. In particular H15 acts transitively

on the15 geometric points of P15, so P15 maps to a rational point on the special fibre of Z. The weaker assertion (iv) suffices below.

All these computations are exact. verify_sage.py independently checks the exported rational inclusions, common mod2 kernel, genera and the quadric calculation used in Section8.3. independent_checks.json records PASS. This independent linear-algebra check is not a separate agent audit.

The curve X has good reduction at2, and H5,H15 have odd order. Their tame quotients Y,Z therefore have good reduction at2, with the quotient maps and diamond actions extending over Z_2. The map f:X -> Z has degree15. Since f_* f^*=[15], its pullback is injective on2-torsion. It follows from (iii) that J_Z(Q)[2]=0 as well.

For an abelian scheme over Z_2, the torsion in the kernel of reduction on Q_2-points is killed by2: doubling carries this kernel into the modulo4 formal subgroup, on which the formal logarithm is injective. Consequently rational torsion reduction is injective on both J_X and J_Z. No assertion of injectivity on all rational points is needed.

8.2. DESCENT TO AN IRREDUCIBLE CUBIC DIVISOR ON Z

Suppose such a D exists. Choose a rational cusp c on X, and put
 h=delta_3. Then
 [hD-D]=(h-1)[D-15c]+15[hc-c]
is rational torsion, by (i) and Manin--Drinfeld. It reduces to zero, by (iv). Hence [hD-D]=0. A nonzero principal divisor hD-D would give a rational function over Q of degree at most15. This is impossible, since gon_Q(X)>=325*(61^2-1)/65536>15. Thus hD=D as divisors.

The order5 group H5 acts freely on characteristic-zero noncuspidal points of X: a diamond stabilizer there can only have order dividing 2 or3, from the extra elliptic automorphisms. Therefore D is the pullback of an irreducible cubic divisor D_Y on Y. Its image D_Z on Z has degree1 or3. Degree1 would give a rational noncuspidal point on X_0(61), forbidden by Mazur’s rational prime-isogeny theorem. Thus D_Z is irreducible of degree3, and f_*D=5D_Z.

Reduction commutes with f and the diamonds. Since delta_2 fixes P15, we obtain 5*(s red(D_Z)-red(D_Z))=0 in the divisor group of the special fibre of Z. That group is torsion-free, so red(D_Z) is s-invariant. (The transitive H15 action above also gives red(D_Z)=3S for one rational point S.) Choose a rational cusp c_Z of Z. Then
 [sD_Z-D_Z]=(s-1)[D_Z-3c_Z]+3[sc_Z-c_Z]
is rational torsion by (ii) and Manin--Drinfeld on the quotient. It reduces to zero, hence [sD_Z-D_Z]=0 by injectivity of torsion reduction on J_Z.

8.3. THE RATIONAL GONALITY OF Z IS AT LEAST4

First C=X_0(61) has no Q-map to P^1 of degree at most3. An exact smooth canonical model, returned by Magma’s SmallModularCurve(61), is a complete intersection of degrees2 and3 in P^3. Its quadric is
 Q=x2^2-x1*x3+x1*x4+x3*x4.
Its symmetric Gram matrix has rank4 and determinant -1/4. Adjunction identifies O(1) with the canonical sheaf, so C is nonhyperelliptic. A rational degree3 pencil would give a rational ruling of this quadric: the canonical spans of the general fibres are lines contained in Q. But the nonsquare discriminant means the two rulings are exchanged

over Q(i), and neither is defined over Q. This excludes a Q-map of degree3. Genus4 excludes degree1. This also agrees with the published classification: Hasegawa--Shimura, Trigonal modular curves, and Jeon, Modular curves with infinitely many cubic points, Theorem2.3. The latter is available at https://doi.org/10.1016/j.jnt.2020.09.006 . Here rational gonality is essential; C is trigonal over Q(i).

Suppose Z had a Q-map u:Z -> P^1 of degree d=2 or3. Since g(Z)=8 exceeds (d-1)^2, its degree-d pencil is unique, by the Castelnuovo--Severi inequality. For these prime degrees, two distinct pencils give a birational map to a bidegree(d,d) curve in P^1 x P^1, which would have genus at most(d-1)^2.

Thus u s=alpha u for alpha in PGL_2(Q), with alpha^2=1. If alpha=1, u factors through C. This is impossible for d=3 by parity, and for d=2 because it would give a degree1 map C -> P^1. If alpha is a nontrivial involution, pass to quotients to obtain a Q-map C -> P^1/<alpha> of degree d. The quotient of P^1 is rational over Q (for example by Luroth’s theorem), so this contradicts the preceding paragraph. Degree1 is excluded by genus. Therefore gon_Q(Z)>=4.

8.4. THE CONTRADICTION

The principal divisor sD_Z-D_Z has positive part of degree at most3. The gonality bound from Section8.3 forces it to be zero. Thus sD_Z=D_Z. The three distinct geometric points of D_Z form one Galois orbit, and s commutes with Galois. All its orbits on those three points have the same length. Since the length divides2 and must divide3, it is1. All three points are fixed by s.

Riemann--Hurwitz for the double cover Z -> C gives exactly

$$(2*g(Z)-2)-2*(2*g(C)-2)=14-12=2$$

geometric fixed points of s in characteristic zero. Three fixed points are impossible. This excludes the last reduction at61 and proves 61 is not in S(15).

9. CHECKING AND SCOPE

verify_auxiliary.py independently checks the156 dyadic matrices, all17 formal-immersion levels and relation bases, the20 known41 cotangent matrices, the gonality53 matrix, the actual rational2-torsion upper bounds, and the16 finite-image inclusions. verify_ranks.py independently checks2578 principal-divisor obstruction matrices. verify_coverage.py checks all prime and local-case ledgers, including the new exclusion of61. quotient61/verify_sage.py independently checks the two new finite-image inclusions, the rational2-torsion bound, the four genera, and the nonsquare canonical-quadric discriminant. The new Magma checks also verify the exact finite-field diamond actions.

The final ledger has status PROVED_WITH_COMPUTER_ASSISTANCE. It accounts for all3551 primes through33131, with17 complete local exclusions and no unresolved primes. The lower and upper sets agree.

No separate agent audit of this S(15) argument is claimed. The matrix checks described above were repeated with independent Sage arithmetic. The new quotient argument closes the previously identified gap at61.

References supplied with the packet:
DKSS, Torsion points on elliptic curves over number fields of small degree,

arXiv:1707.00364, especially the rank-zero primes, rational-torsion annihilators, formal-immersion criterion and Hecke intersection formula.
Derickx--Stoll, Prime order torsion on elliptic curves over number fields, Part I: Asymptotics, arXiv:2505.14109v1, Table2, Proposition4.1, Remark4.4, Corollary7.5 and the gonality bound.
The prior dyadic theorem and van Hoeij’s explicit occurrence data.

Additional reference for Section8:
Y. Hasegawa and M. Shimura, Trigonal modular curves, Acta Arithmetica88 (1999); D. Jeon, Modular curves with infinitely many cubic points, Journal of Number Theory219 (2021),344-355, Theorem2.3, https://doi.org/10.1016/j.jnt.2020.09.006 .

CORRECTION TO THE ORDINARY LEVEL97 CERTIFICATES (29 October 2026)

The original code polynomial sigma^6-sigma^4-sigma^2+1 does not annihilate the strange order12 character. It has been replaced by sigma^12-1 and, on the three closed orbits of size24 in degrees10/14, (sigma-1)*Phi12(sigma). All ordinary level97 residue orbits through degree17 have been recomputed with the corrected map and independently checked in Sage. The relevant lower-degree matrices here are exact subsets of that universal certificate. The other level97 arguments are unchanged.
See outputs/S97_correction.pdf and work/degree17/full97/sieve.json.